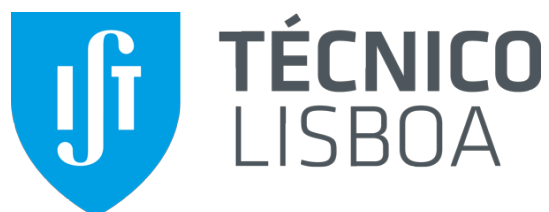


Universidade de Lisboa
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Project in Aerospace Engineering



**Analysis of Stabilized Approaches at Lisbon
Airport**

by

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1 Introduction

The continuous growth of global air traffic places immense pressure on airport infrastructure, demanding operational efficiency without compromising safety standards. Single-runway systems, such as Humberto Delgado Airport (LPPT) in Lisbon, operate in highly constrained environments where arrival sequencing and runway occupancy times are tightly managed. In such high-density scenarios, the execution of a Stabilized Approach (SA) is universally recognized by aviation authorities as a critical safety barrier against approach and landing accidents, specifically Runway Excursions (RE) and Loss of Control In-flight (LOC-I).

A stabilized approach is characterized by maintaining a constant glide path, tracking the runway centerline, and managing aircraft energy within narrow margins before reaching a critical altitude, typically the 1000 ft Above Ground Level (AGL) gate for Instrument Meteorological Conditions (IMC). While standard operating procedures dictate strict binary thresholds for these parameters, the real-world execution is influenced by atmospheric conditions, air traffic control (ATC) vectoring, and pilot energy management. Consequently, accurately measuring fleet compliance requires shifting from theoretical predictions to empirical flight path monitoring.

1.1 Motivation

Historically, predictive safety models have relied heavily on theoretical kinematic models derived from the Flight Management System (FMS). However, these on-board models often fail to account for the tactical realities of a flight's progress, leading to discrepancies between assumed performance and actual operations. The primary motivation of this project is to bridge that gap by utilizing real-world historical tracking to assess operational risk dynamically.

This assessment is made possible by the availability of high-fidelity multi-sensor surveillance data provided by NAV Portugal. By fusing Radar, Automatic Dependent Surveillance-Broadcast (ADS-B), and Multilateration (MLAT) sources, the dataset captures comprehensive telemetry for all airspace users with minimal data gaps. Utilizing this ANSP-grade (Air Navigation Service Provider) data shifts the scope of the project from a generic data analysis exercise to a highly realistic operational safety assessment, mirroring the diagnostic capabilities of modern aviation safety departments.

Furthermore, traditional safety metrics often rely on rigid, binary limits that create artificial performance cliffs and generate false positives due to unknown variables like landing weight. This project is motivated by the need to implement adaptive, data-driven parameters and continuous evaluation metrics that better reflect the nuanced operational realities of different aircraft categories.

1.2 Project Objectives

The overarching goal of this report is to quantify, characterize, and predict unstabilized approaches at LPPT using historical ATC surveillance track data. To achieve this, the following specific objectives have been established:

- **Data Pipeline Engineering:** To process, clean, and calibrate raw multi-source surveillance data, integrating local METAR weather conditions and applying zero-offset calibrations to correct systematic spatial biases.
- **Data-Driven Gate Evaluation:** To evaluate multi-dimensional stabilization criteria at the 1000 ft AGL gate. This includes implementing a dynamic, data-driven approach speed (V_{REF}) extracted from the median airspeed within the stabilized 3 NM to 1 NM window, thereby bypassing the limitations of theoretical weight assumptions.
- **Continuous Stability Scoring:** To overcome the limitations of binary pass/fail criteria by developing a Continuous Stability Score. This model will utilize a Gaussian penalty function, $S = \exp(-x^2/2\sigma^2)$, to assign an objective quality metric to each approach, allowing for accurate fleet benchmarking.
- **Predictive Nowcasting:** To assess the feasibility of early-warning predictors by analyzing the spatial and energy states of aircraft at the 6 NM mark, determining if early deviations can reliably predict late-stage instability at the 1000 ft gate.

2 Data Processing

The foundation of this safety assessment relies on high-fidelity surveillance tracking provided by the official Air Navigation Service Provider [3], NAV Portugal, corresponding to two complete different days: 2023/05/01 and 2024/06/17. The

raw dataset comprises over 150 individual spreadsheet files, containing hundreds of thousands of telemetry rows captured during active operations at the airport. Utilizing this multi-sensor data fusion guarantees an exceptional level of completeness and accuracy, minimizing track gaps and elevating the assessment to an industrial operational standard.



Figure 1: Lisbon Airport radar station.

2.1 Raw Data Structure

To facilitate programmatic handling, the massive volume of initial spreadsheet files was standardized into Comma-Separated Values (CSV) format before initiating the analytical pipeline. The raw telemetry provided a comprehensive snapshot of every aircraft’s state at any given radar sweep, including essential identification features such as flight callsigns, unique transponder codes, and ICAO aircraft type designators. Furthermore, the dataset contained critical spatial and kinematic columns, namely timestamps, latitude and longitude coordinates, Mode-S pressure altitude, ground speed, and track angle, which served as the baseline vectors for all subsequent trajectory reconstructions [1].

	Latitude (°)	Longitude (°)	Column1.geometry.type	Column1.type	Column1.properties.Measured flight level (int)	Column1.properties.Mode3A (String)	Column1.properties.ADEP
0	37.134980	-9.051898	Point	Feature	330.0	2626.0	LEMD
1	37.520880	-8.723236	Point	Feature	350.0	2610.0	LEMD
2	37.678868	-7.752367	Point	Feature	350.0	3757.0	LEMD
3	37.917826	-8.832922	Point	Feature	300.0	3334.0	LPPT
4	37.858211	-9.626250	Point	Feature	350.0	2617.0	LEMD

Figure 2: Raw data, part 1.

Column1.properties.AType	Column1.properties.Track number (int):	vx (m/s)	vy (m/s)	Column1.properties.WTC (H/M/L)	Column1.properties.Callsign (String)	Column1.properties.Time update (HH:mm:ss.SSS):
A332	2134	-133.50	-196.75	H	IBEF011	23:23:55.796
B789	2140	-139.00	-203.25	H	AEA057	23:23:55.796
B789	2145	-66.50	-250.00	H	AEA041	23:23:56.093
B734	2179	48.75	-224.25	M	TAY1MN	23:23:56.093
A359	2136	-194.25	-143.50	H	IBEF833	23:23:56.093

Figure 3: Raw data, part 2.

2.2 Spatial Filtering and Arrival Identification

The surveillance radar inherently captures all air traffic traversing its coverage volume, which includes high-altitude overflights and en-route transit traffic entirely unrelated to Humberto Delgado Airport. Therefore, a rigorous geographic and operational filtering phase was essential to isolate the relevant dataset. The pipeline was configured to first constrain the telemetry to flights operating strictly within the LPPT terminal area, and subsequently, a vertical trajectory filter was applied to isolate only the arriving aircraft, systematically discarding all departures and unrelated traffic.

2.3 Flight Trajectory Reconstruction

A significant challenge associated with raw, continuous multi-sensor surveillance is the fragmentation of trajectories and the repetition of standard callsigns over different days. To ensure data integrity, the algorithm was engineered to assign a completely new, unique flight identifier (flight-id) to every individual arrival operation. This prevented distinct flights that happened to share the same commercial callsign across different dates from being erroneously merged into a single trajectory. Once these unique operations were defined, duplicate radar returns were rigorously removed to prevent statistical distortion in the downstream calculations.

2.4 Code Availability and Reproducibility

All data handling, spatial filtering, and duplicate removal processes were executed within a Python environment using advanced dataframe manipulation techniques. To maintain the readability of this main report, no raw source code is included within the body chapters. The complete Python scripts utilized for this initial data processing phase, as well as all subsequent algorithms, stability scoring evaluations, and plotting functions mentioned throughout the remainder of this document, are fully compiled and available for technical review in Appendix A.

3 Stabilization Criteria

3.1 The 1000 ft AGL Stabilization Gate

The core of this safety assessment relies on defining an objective, measurable point during the final approach to evaluate the operational state of the aircraft. In accordance with standard aviation safety guidelines for Instrument Meteorological Conditions (IMC), the 1000 ft Above Ground Level (AGL) gate was selected as the strict stabilization threshold. At this exact altitude, the aircraft must be established on the correct lateral and vertical flight paths, sinking at an acceptable rate, and managing energy within safe tolerances. To achieve a comprehensive evaluation, the algorithmic criteria were divided into spatial, vertical, and kinetic dimensions [4].

3.2 Spatial Parameters

The spatial trajectory of the aircraft determines its alignment with the runway threshold. To classify an approach as spatially stable, the telemetry was evaluated against two primary angular envelopes:

- **Vertical Glide Slope:** The aircraft must maintain the standard 3.0° descent path required for runway at LPPT. The stabilization algorithm allows for a maximum vertical deviation of $\pm 0.5^\circ$. Any trajectory dropping below 2.5° or floating above 3.5° at the 1000 ft gate triggers an unstable path alert.
- **Lateral Localizer Tracking:** The aircraft must be aligned with the extended runway centerline. A maximum lateral deviation of $\pm 5.0^\circ$ from the true tracking course is permitted to absorb minor navigational corrections or crosswind adjustments.

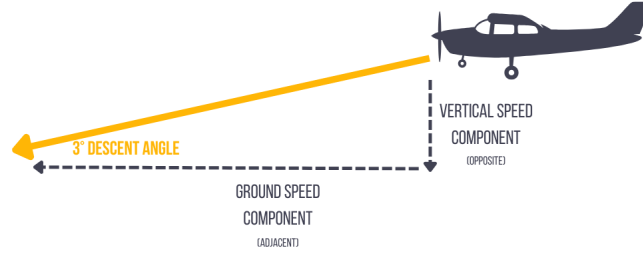


Figure 4: Vertical glide slope.

3.3 Descent Profile: Vertical Rate

High sink rates near the ground are leading indicators of hard landings and structural stress. The kinematic criteria enforce a strict vertical rate limit. To be considered stable, the aircraft's descent rate—derived from the filtered altitude changes over time—must not exceed -1000 feet per minute (fpm). Sink rates faster than this threshold at the 1000 ft mark indicate that the aircraft is "diving" to capture the glide path, thereby failing the stabilization gate.

3.4 Energy Management

Energy management is evaluated by comparing the aircraft's Indicated Airspeed (IAS) against its target final approach speed, commonly referred to as the reference speed (V_{REF}). To satisfy the stabilization criteria at the 1000 ft gate, the kinetic energy of the aircraft must be tightly controlled to prevent runway overruns or aerodynamic stalls. The established operational tolerance mandates that the aircraft must not cross the gate at a speed exceeding +20 knots or falling below -10 knots of its specific V_{REF} . Speeds outside this envelope indicate a high-energy or low-energy state, both of which trigger an unstabilized approach classification.

3.5 Stability Scoring Model

While binary thresholds (Stable vs. Unstable) are standard in the industry, they create artificial analytical cliffs where negligible differences—such as a descent rate of -999 fpm versus -1001 fpm—drastically alter the safety classification. To provide a more nuanced operational assessment, a Continuous Stability Score was engineered using a Gaussian penalty function. The performance of each flight is evaluated on

a continuous scale from 0 to 100 using the following formula:

$$S = \exp\left(-\frac{x^2}{2\sigma^2}\right)$$

In this model, x represents the actual deviation from the ideal target (e.g., the exact knots above V_{REF} or the degrees off the centerline), and σ represents the accepted operational tolerance. By aggregating the Gaussian scores for speed, lateral track, and vertical rate, the model moves beyond simple pass/fail metrics, enabling precise ranking of fleet performance and deeper insights into approach quality.

4 Algorithmic Calibrations and Kinematic Modeling

To extract accurate flight kinematics from the raw surveillance signals, several mathematical transformations were required. The complete Python source code for all the algorithmic calibrations detailed in this chapter is fully documented and available in Appendix A.

4.1 Meteorological Altitude Correction (QNH)

Raw surveillance data transmits altitude as Mode-S pressure altitude, which is referenced to the standard atmospheric pressure setting (1013.25 hPa). To accurately determine the aircraft’s true vertical position relative to the ground, this raw data must be calibrated using actual local weather conditions. Historical METAR data for Humberto Delgado Airport was integrated into the pipeline to extract the specific QNH (barometric pressure adjusted to sea level) at the time of each approach. Applying this barometric correction translates the standard pressure altitude into a true Above Ground Level (AGL) altitude, ensuring that the 1000 ft stabilization gate is evaluated precisely at the correct physical elevation.

4.2 Noise Reduction

Radar and multilateration systems inherently introduce high-frequency noise and quantization artifacts into the positional telemetry. Calculating instantaneous rates of change, such as the Vertical Rate (fpm), directly from raw altitude differences

between radar sweeps results in highly erratic and physically impossible spikes.

To resolve this kinematic distortion, a Savitzky-Golay filter was applied to the time-series data. This digital polynomial smoothing filter was selected because it successfully smooths the altitude and spatial telemetry while preserving the shape, peaks, and underlying signal profile of the aircraft’s trajectory. This ensures that the descent rates evaluated at the 1000 ft gate are realistic aerodynamic measurements rather than radar-induced anomalies.

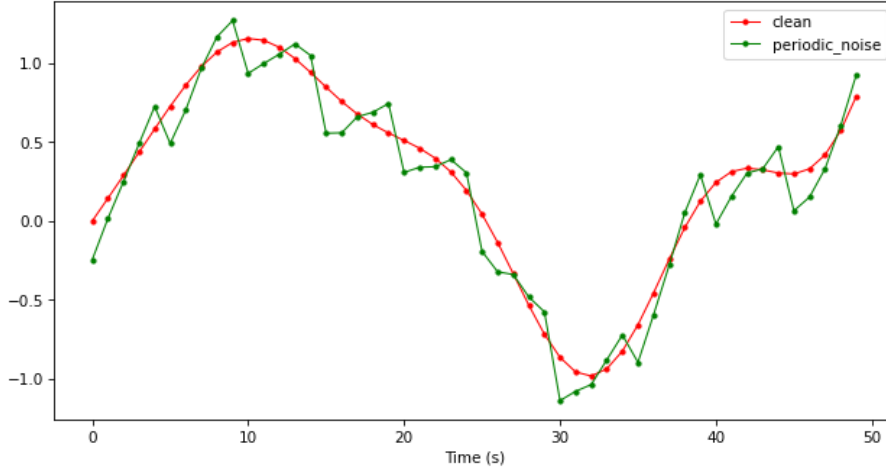


Figure 5: Example of noise smoothing with Savitzky-Golay filter.

4.3 Data-Driven Reference Speed (V_{REF}) Estimation

A major limitation in surveillance-based safety assessments is the absence of aircraft weight telemetry, making it impossible to calculate a theoretical final approach speed (V_{REF}) using standard Flight Management System formulas. Furthermore, radar networks primarily track Ground Speed, whereas aerodynamic energy management criteria rely strictly on Indicated Airspeed (IAS).

To resolve this discrepancy, historical wind data[6] from both operational days covered in the dataset was integrated into the kinematic pipeline. By applying the specific wind components to the flight tracks, the algorithm successfully converted the radar-derived Ground Speed into an accurate IAS.

With the true airspeed calculated, an empirical, data-driven methodology was engineered to estimate the V_{REF} . The algorithm calculates a unique baseline speed

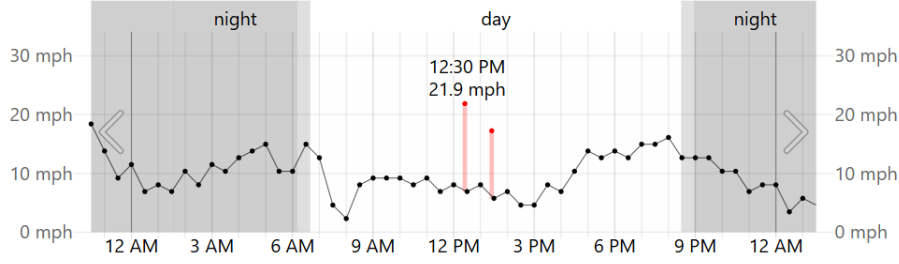


Figure 6: Windspeed on 2023/05/01 in Lisbon.

for each individual flight by monitoring its IAS during the most stable, late-stage segment of the final approach—specifically the geometric window between 3 NM and 1 NM—from the runway threshold. By extracting the median IAS within this window, the model establishes a customized V_{REF} baseline. This purely empirical approach naturally absorbs unknown operational variables, such as landing weight, local wind gradients, and specific flap configurations, providing a highly reliable target for the energy management evaluation.

5 Global Stability Overview

After applying the multi-dimensional stabilization criteria to the filtered surveillance data, a total of 576 arrivals were successfully evaluated at the 1000 ft AGL gate. The global evaluation reveals a high operational compliance rate, with 89.24% of the fleet crossing the gate within all established safety limits. Consequently, the global unstabilized approach rate stands at 10.76%.

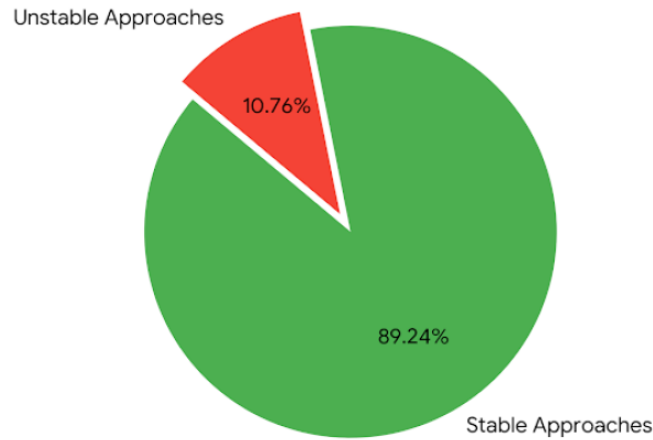


Figure 7: Global stability overview.

This unstabilized rate is a highly realistic metric for a congested, single-runway airport environment like Humberto Delgado (LPPT). It confirms the efficacy of the data engineering pipeline, particularly the Savitzky-Golay filtering and the data-driven V_{REF} estimation. By dynamically adjusting the baseline speeds to actual flight behavior, the algorithm successfully filtered out the massive false-positive anomalies that typically plague raw radar safety assessments, leaving a mathematically sound representation of true operational risk.

5.1 Instability Breakdown by Criteria

Analyzing the specific triggers of instability provides critical insight into the operational pressures driving approach deviations at LPPT. It is important to note that the sum of the individual criteria failure rates exceeds the total global unstabilized rate (10.76%). This statistical overlap indicates that multiple flights failed more than one parameter simultaneously—for instance, carrying excess energy while also exceeding the descent rate limit to capture the glide path.

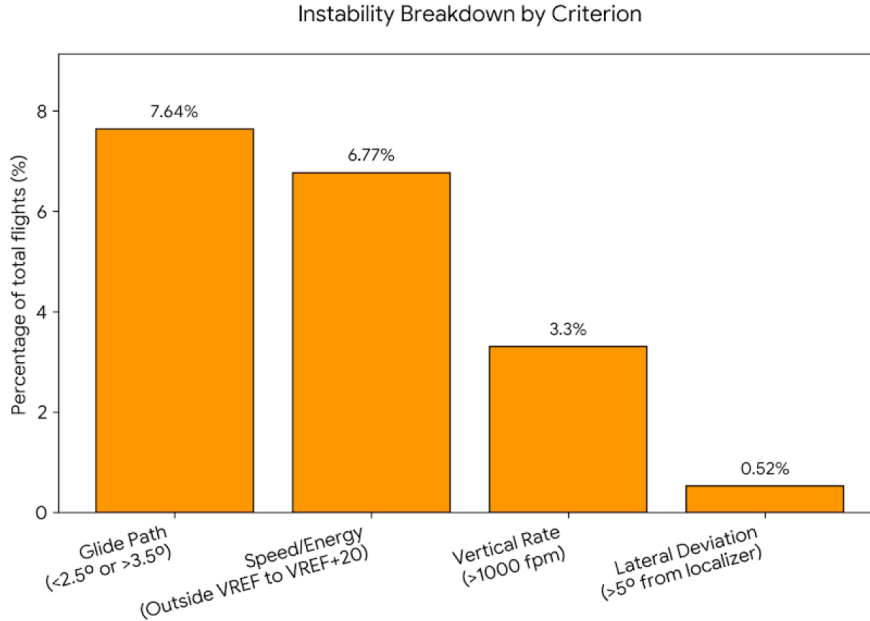


Figure 8: Instability breakdown by criteria.

The breakdown of the individual parameters reveals distinct operational trends:

- **Glide Path Deviation (7.64%):** Vertical trajectory deviations (falling below 2.5° or floating above 3.5°) represent the highest single cause of instability.

This is a common symptom in high-density terminal areas where Air Traffic Control (ATC) may issue tight vectoring or late descent clearances, forcing crews to capture the glide slope from above or adjust their vertical profile late in the approach.

- Speed and Energy Management (6.77%): The second leading trigger occurred when aircraft crossed the 1000 ft gate carrying too much kinetic energy, busting the upper tolerance of $V_{REF} + 20$ knots. This strongly correlates with standard ATC sequencing practices that often request crews to maintain high speeds until the outer marker to maximize runway throughput.
- Vertical Rate Exceedances (3.30%): Flights descending faster than -1000 fpm account for a smaller, yet critical, portion of the unstabilized pool. These exceedances are often kinematically linked to the glide path and energy issues, as aircraft increase their sink rate to bleed off altitude and capture the nominal 3.0° cone.
- Lateral Deviation (0.52%): Lateral instability is virtually non-existent in this sample. This aligns with modern aviation capabilities, as the lateral localizer is typically captured early in the approach sequence, and the autopilot or flight director maintains centerline tracking with extremely high precision.

6 Analysis by Aircraft Category

To extract meaningful operational insights, analyzing the fleet as a single homogeneous group is insufficient, as different aircraft sizes possess vastly different aerodynamic characteristics and approach speeds. Therefore, the surveillance dataset was segmented into distinct operational families.

In this case, the algorithm extracted the specific ICAO Aircraft Type Designator (e.g., A320, B77W, E190, AT72) transmitted by each aircraft’s Mode-S transponder. A custom classification dictionary was engineered within the Python pipeline to map these raw ICAO codes into five macro categories: Narrowbody, Widebody, Regional, Turboprop, and Other (which includes business jets and non-standard general aviation). This segmentation allows for a direct comparison of how structural weight classes influence stabilization performance at the 1000 ft gate.

6.1 Categorical Stability Performance

Applying the stability criteria across the segmented fleet reveals significant performance variations among the different aircraft categories. The instability rates per family are distributed as follows:

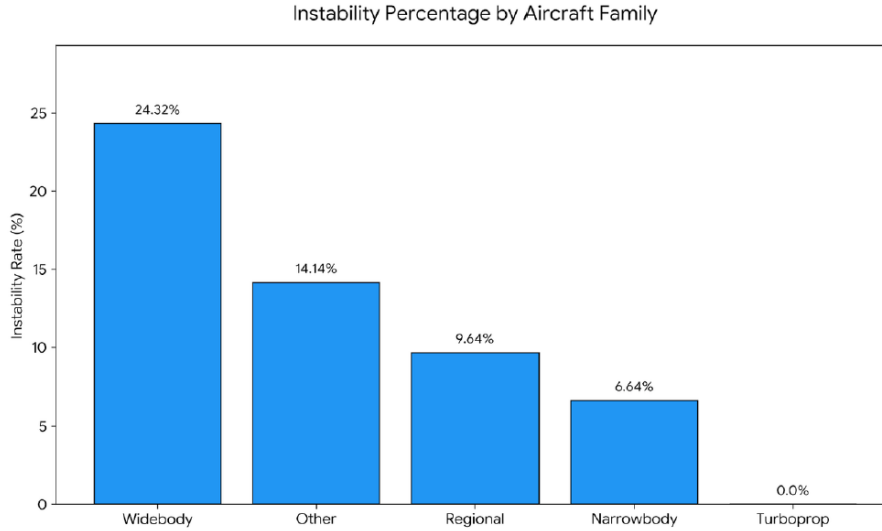


Figure 9: Instability percentage by aircraft family.

The data clearly indicates that lighter, propeller-driven aircraft (Turboprops) easily comply with the stabilization gate, achieving a perfect 100% stability rate. This is aerodynamically expected, as the large, unfeathered propeller blades of a turboprop at flight idle create massive aerodynamic drag, allowing the aircraft to decelerate and descend simultaneously with minimal effort. Conversely, the instability rate rises progressively with heavier jet aircraft.

6.2 The Widebody Energy Management Challenge

The most critical finding from this categorical analysis is the pronounced instability rate of Widebody aircraft, with nearly one in four (24.32%) failing to fully stabilize by the 1000 ft gate. This deviation is deeply rooted in their physical and aerodynamic characteristics. Heavy jets possess immense mass, high inertia, and very "clean" aerodynamic designs (low drag), making it exceptionally difficult for them to decelerate and descend simultaneously.

In a congested single-runway environment like LPPT, ATC frequently requests aircraft to maintain high speeds to maximize arrival throughput. While lighter

jets can quickly shed this kinetic energy just before the gate using speed brakes and gear, Widebodies require a significantly longer spatial window to decelerate. Consequently, when subjected to the same tight ATC sequencing as lighter traffic, heavy jets frequently cross the 1000 ft mark still carrying excess speed.

7 Extended Analysis

7.1 Stability Scoring Results

While binary pass/fail metrics provide a clear safety boundary, they do not measure the overall quality or precision of an approach. Applying the continuous Gaussian scoring model—which evaluates flights on a scale of 0 to 100 based on their proximity to the ideal nominal profile—yields a more nuanced ranking of the fleet’s performance:

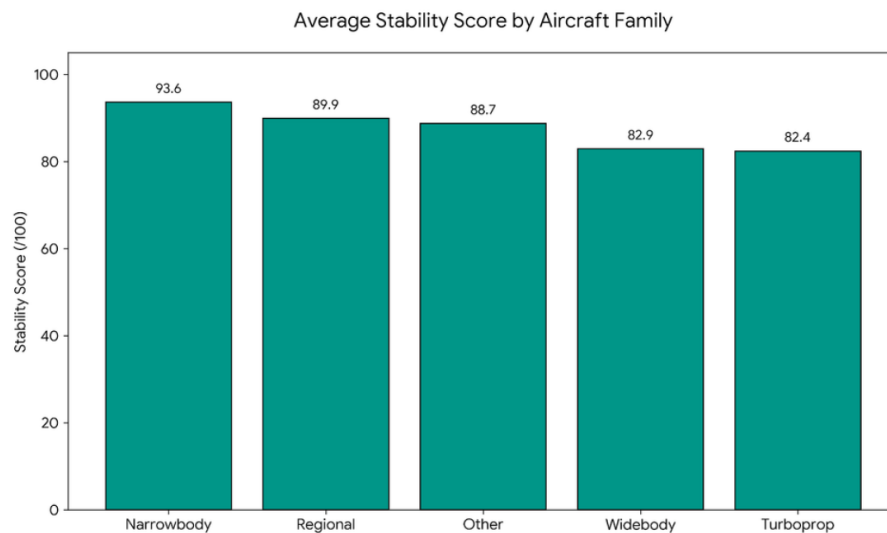


Figure 10: Average stability score by aircraft family.

Narrowbody aircraft achieve the highest scores, indicating that they not only remain within safe boundaries but also adhere very closely to the mathematically ideal 3.0° glide path and optimal descent rates. As expected, Widebodies score lower, reflecting their struggle to manage energy and maintain the centerline with the same agility.

7.2 The Turboprop Paradox: Safety vs. Precision

A notable anomaly in the data is the performance of the Turboprop category. In the binary evaluation, Turboprops achieved a perfect 0.0% instability rate, yet they recorded the lowest continuous score in the fleet (82.4 / 100).

This apparent paradox highlights the fundamental difference between operational safety and mathematical precision. The binary metric simply confirms that no Turboprop crossed the critical red lines (e.g., they never exceeded +20 knots or sank faster than -1000 fpm). However, the Gaussian scoring function penalizes any deviation from the absolute ideal center. Turboprops, due to their distinct aerodynamic characteristics and slower operational speeds, naturally fly different descent profiles. While they are intrinsically safe and never breach the dangerous outer limits, they consistently operate slightly off the strict "ideal" jet profile. These continuous minor deviations accumulate as penalty deductions in the Gaussian model, resulting in a lower overall score despite a flawless safety record.

7.3 Predictive Nowcasting

The ultimate goal of analyzing surveillance data is not merely retrospective auditing, but proactive risk management. To assess predictive capabilities, the algorithm evaluated the spatial and energy states of the fleet at an early-warning window: exactly 6 Nautical Miles (NM) from the runway threshold [5].

The objective was to determine if positional errors at 6 NM strongly correlate with eventual unstabilized approaches at the 1000 ft gate (approximately 3 NM later). By plotting the lateral and vertical deviations at the 6 NM mark, the methodology successfully mapped a predictive boundary.

The plotted data[8.2] clearly demonstrates that the 6 NM window serves as a highly reliable early indicator. Aircraft that stray outside the central geometric safe zone at this early stage have a statistically high probability of failing the stabilization criteria at the 1000 ft gate. This confirms that utilizing continuous surveillance tracking can effectively support Air Traffic Control by providing predictive nowcasting alerts long before an aircraft reaches the critical final approach phase.

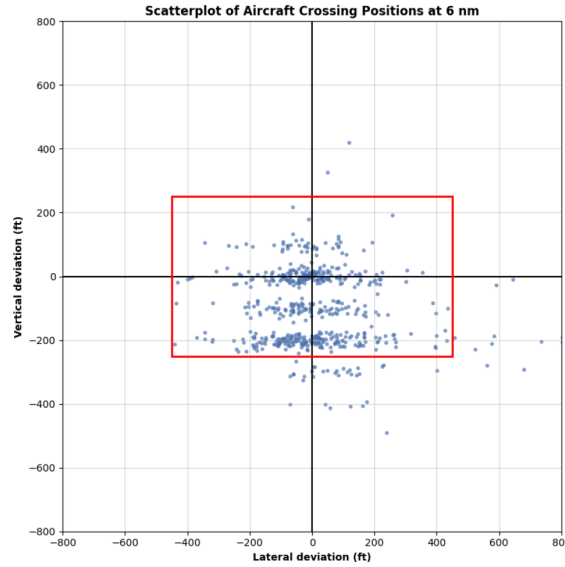


Figure 11: Aircraft crossing positions at 6NM.

8 Case Studies

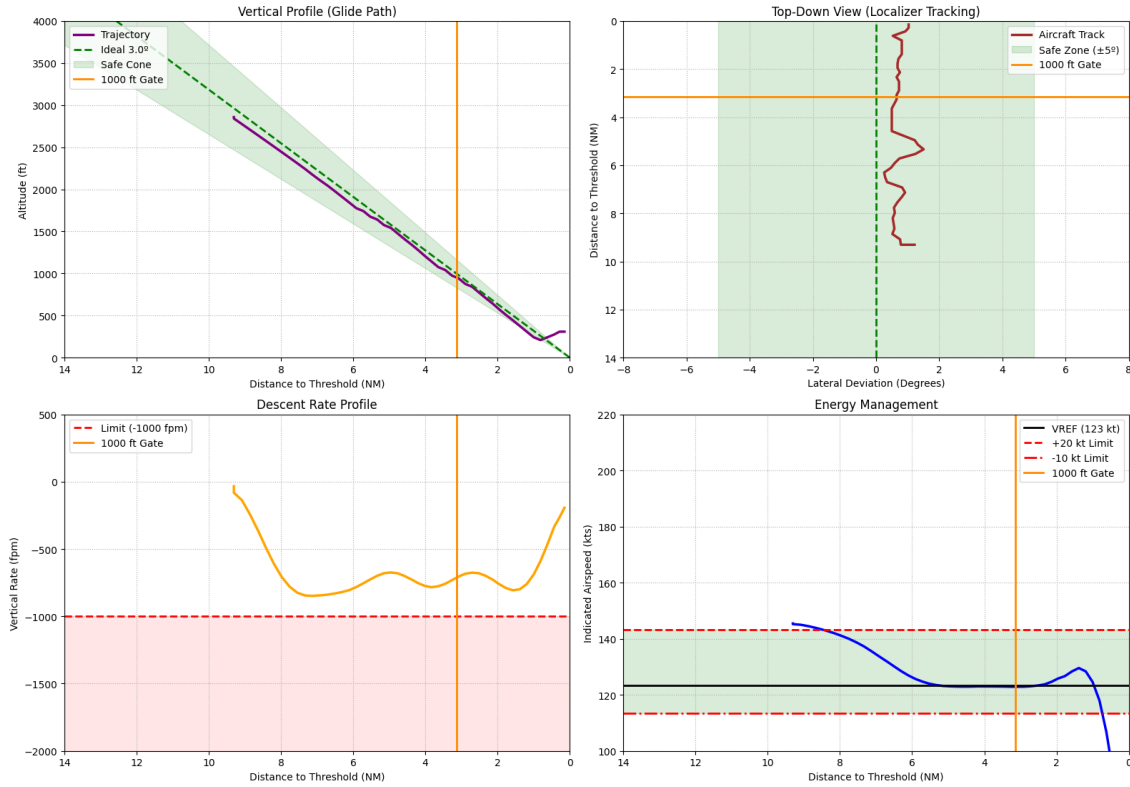
To bridge the gap between statistical aggregates and individual flight dynamics, this chapter presents two distinct operational profiles captured by the surveillance network on June 17, 2024. The telemetry for each flight is visualized through a unified 4-axis dashboard, aligning the lateral track, vertical glide path, descent rate, and energy management strictly by distance to the runway threshold[2].

8.1 Stable Flight Profile (EJU49HF)

Flight EJU49HF represents a nominal, fully stabilized approach. As observed in the accompanying dashboard, the aircraft demonstrates a textbook convergence into all safety envelopes prior to the 1000 ft AGL gate.

The spatial trajectory graphs show the aircraft centered within the 3.0° vertical cone and the extended centerline localizer. Concurrently, the kinematic plots confirm that the descent rate is steadily maintained within safe limits (avoiding the -1000 fpm boundary). Crucially, the energy management plot displays a smooth, progressive deceleration, allowing the aircraft to cross the stabilization gate perfectly matched to its dynamically calculated V_{REF} .

1. STABLE APPROACH Flight ID: EJU49HF_2024-06-17_1.0



Note: The minor data anomalies visible in the graphs approaching 0 NM are standard surveillance artifacts caused by ground clutter and signal multipath near the runway surface. Because these sensor fluctuations occur well below the 1000 ft evaluation gate, they have no impact on the accuracy of the stabilization assessment.

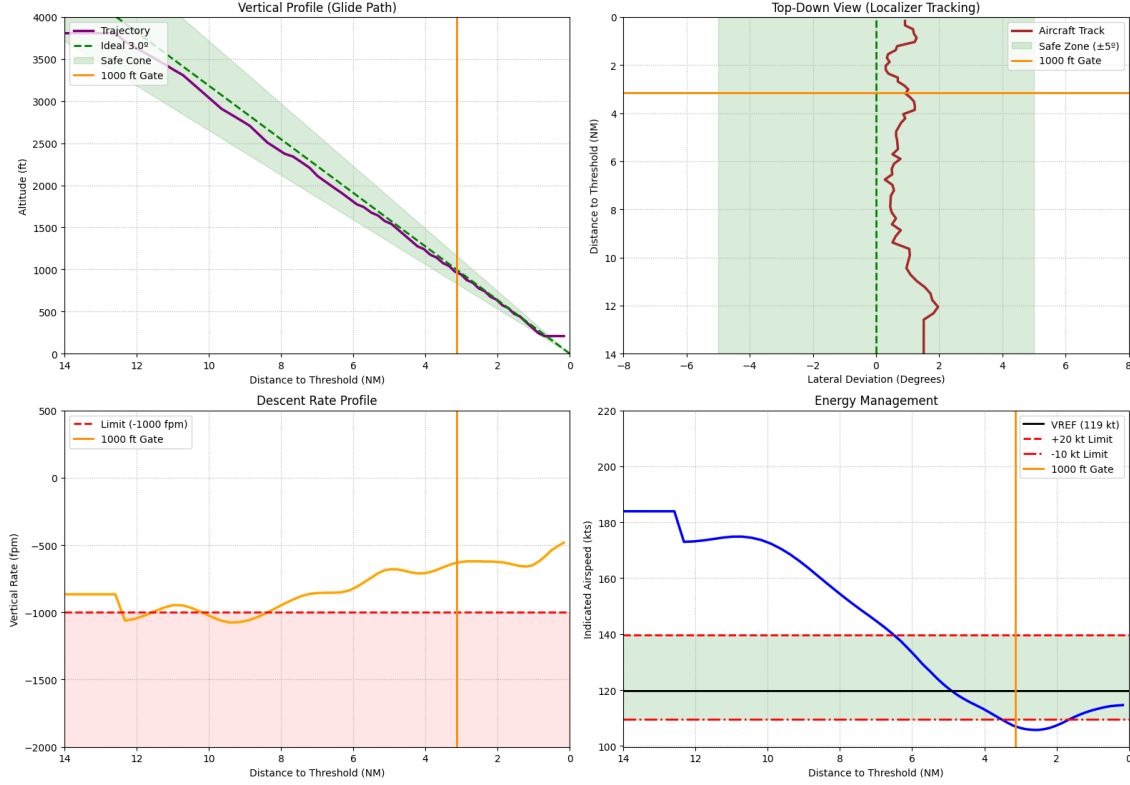
8.2 Unstable Flight Profile (TCV606)

Flight TCV606 illustrates a highly deceptive operational risk: an approach that is spatially flawless but kinetically unstable.

Reviewing the top spatial graphs, the aircraft captures the lateral localizer and vertical glide path with absolute precision. Furthermore, the vertical rate remains compliant throughout the descent. Based solely on its geometric flight path, this approach appears perfectly safe.

However, the energy management plot located in the bottom right reveals a critical operational anomaly. The aircraft fails to shed kinetic energy in time, crossing the 1000 ft gate carrying excessive speed that sharply busts the upper tolerance

2. HIGH ENERGY INSTABILITY (Speed Bust) Flight ID: TCV606_2024-06-17_1.0



limit of $V_{REF} + 20$ knots. This specific case study vividly reinforces the necessity of a multi-dimensional assessment framework; relying solely on spatial tracking would have erroneously classified a high-risk, high-energy approach as completely safe.

9 Conclusions

This project successfully transitioned the assessment of Stabilized Approaches at Humberto Delgado Airport (LPPT) from theoretical frameworks to a highly realistic, data-driven operational reality. By engineering a robust Python pipeline to process and calibrate multi-sensor surveillance data provided by NAV Portugal, this study demonstrated the immense value of using high-fidelity historical tracking for aviation safety management.

The evaluation of the fleet at the 1000 ft AGL gate yielded several critical operational insights:

- **Realistic Risk Baseline:** The global unstabilized approach rate was determined to be 10.76%. The primary triggers for these deviations were glide path capture

issues and excess kinetic energy, reflecting the intense operational pressures and tight vectoring inherent to a congested, single-runway environment.

- The Widebody Energy Challenge: Fleet segmentation revealed a stark contrast in aerodynamic performance. While lighter aircraft easily complied with stabilization gates, Widebody aircraft recorded a 24.32% instability rate. This highlights a systemic challenge: heavy, aerodynamically clean jets struggle to simultaneously decelerate and descend when subjected to the same high-speed ATC sequencing as narrower traffic.

Beyond statistical auditing, this project introduced significant methodological innovations to standard safety practices, like the Data-Driven V_{REF} and Continuous Scoring. Bypassing the limitations of unknown landing weights, the algorithm successfully estimated individual approach speeds using empirical late-stage telemetry. Furthermore, the implementation of a continuous Gaussian scoring function proved that evaluating the quality of an approach provides far deeper insights than binary pass/fail limits. The "Turboprop Paradox"—where propeller aircraft were 100% safe but scored lowest in geometric precision—perfectly illustrates the nuance captured by this continuous model.

In conclusion, this project validates that integrating advanced data engineering with ANSP-grade surveillance not only provides an accurate diagnostic tool for historical safety compliance, but also establishes the foundation for proactive, predictive risk management in modern air traffic operations.

10 Appendix A: Python Scripts

The following Python scripts represent the core data engineering and kinematic modeling pipeline utilized to process the NAV Portugal surveillance data and calculate the stability statistics.

A Filtering arrivals

```
1 import pandas as pd
2
3 print("Filtering arrivals to Lisbon (LPPT)...")
4
5 # Asumiendo que tu dataframe original gigante se llama 'df'
6 df_llegadas = df_limpio[df_limpio['ades'] == 'LPPT'].copy()
7
8 print(f"Total arrival radar pings: {len(df_llegadas)}")
```

B Calculating vertical rate

```
1 print("Calculating robust raw Vertical
   Rate (fpm)...")
2
3 # Al usar periods=5 comparamos cada
   lectura con la de 5 pings atr s.
4 # Esto puentea el "efecto escalera" de
   la cuantizaci n del radar Mode S.
5 dt = df_llegadas.groupby('flight_id')['
   time'].diff(periods=5).dt.
   total_seconds()
6 dz = df_llegadas.groupby('flight_id')['
   altitud_ft'].diff(periods=5)
7
8 # Tasa de descenso media de la ventana
   en pies por minuto
9 df_llegadas['vertical_rate_fpm'] = (dz /
   dt) * 60
10
11 # Las primeras 5 filas de cada vuelo
   dar n NaN al no tener 5 previas.
12 # Como corresponden al inicio del
   descenso a FL300 y a nosotros nos
   importa la puerta de 1000 ft,
13 # rellenamos con 0 de forma segura.
14 df_llegadas['vertical_rate_fpm'] =
   df_llegadas['vertical_rate_fpm'].
   fillna(0)
15
16 # Limpieza estricta: si por alg n
   microcorte dt es exactamente 0,
   evitamos infinitos matem ticos
17 df_llegadas['vertical_rate_fpm'] =
   df_llegadas['vertical_rate_fpm'].
   replace([float('inf'), float('-inf')
   ], 0)
18
19 print("Robust Vertical Rate calculated
   successfully.")
```

C QNH correction

```
1 # --- BLOCK 4: QNH CORRECTION ---
2 print("Applying barometric correction (
   QNH) based on historical weather..."
   )
3
4 # Presiones pasadas a hPa (milibares)
5 qnh_2023 = 30.04 * 33.8639
6 qnh_2024 = 29.93 * 33.8639
```

```

7 qnh_std = 1013.25
8
9 # Mapeamos la presi n seg n el a o
  del vuelo
10 df_llegadas['QNH_day'] = np.where(
    df_llegadas['time'].dt.year == 2023,
    qnh_2023, qnh_2024)
11
12 # Aplicamos la f rmula f sica
13 df_llegadas['altitud_ft_corr'] =
    df_llegadas['altitud_ft'] + 27.3 * (
    df_llegadas['QNH_day'] - qnh_std)
14
15 print("True altitude computed
    successfully.")

```

D Savitzky-Golay filter

```

1 # --- BLOCK 5: SAVITZKY-GOLAY FILTER ---
2 from scipy.signal import savgol_filter
3
4 print("Applying Savitzky-Golay advanced
    smoothing filter to descent profiles
    ...")
5
6 def apply_savgol(data):
7     # Usamos una ventana de 11 lecturas
    para absorber el ruido y evitar
    picos de 30k fpm
8     if len(data) < 11:
9         return data
10    return savgol_filter(data,
        window_length=11, polyorder=2)
11
12 # Aplicamos agrupando de forma estricta
    por el identificador nico limpio
13 df_llegadas['savgol_vr_fpm'] =
    df_llegadas.groupby('flight_id')['
    vertical_rate_fpm'].transform(
    apply_savgol)
14
15 print("Descent profiles smoothed.")
16
17 # Verificaci n visual comparativa
18 savgol_check = df_llegadas[(df_llegadas[
    'callsign'] == 'ACA812') &
    (df_llegadas[
    'altitud_ft_corr'] <= 1500) &
    (df_llegadas[
    'altitud_ft_corr'] >= 800)]
19
20
21
22 print("\nComparison at Gate Approach (
    Raw vs. Corrected Altitude &
    Smoothed VR):")
23 display(savgol_check[['flight_id', 'time'
    ], 'altitud_ft', 'altitud_ft_corr',
    'vertical_rate_fpm', 'savgol_vr_fpm'
    ]).head(8))
24
25 # Criterio: Tasa de descenso no superior
    a 1000 fpm
26 # Como nuestro avi n desciende, el
    valor es negativo, as que buscamos
    valores menores a -1000
27 df_llegadas['vr_unstable'] = df_llegadas
    ['savgol_vr_fpm'] < -1000

```

E V_{REF} / IAS calculations

```

1 # --- BLOCK 6: PROFILING ENERGETICO
    FINAL (VIENTO + VREF EMP RICA DATA-
    DRIVEN) ---
2 import pandas as pd
3 import numpy as np
4 from scipy.signal import savgol_filter
5
6 print("Step 6.1: Procesando METAR y
    filtrando ruido de radar...")
7
8 # Limpieza de memoria
9 cols_to_drop = [
    'wind_dir', 'wind_speed', '
    ground_speed_kts', 'gs_smooth_kts',
10    'headwind_kts', 'tas_est_kts', '
    ias_est_kts', 'dynamic_vref', '
    speed_unstable']
11
12 df_llegadas = df_llegadas.drop(columns=[
    col for col in cols_to_drop if col
    in df_llegadas.columns], errors='
    ignore')
13

```

```

14 # 1. INTEGRACIÓN DE METAR
15 weather_data = [
16     ('2023-05-01 00:00:00', 340, 7), ('
17     '2023-05-01 01:00:00', 340, 9),
18     ('2023-05-01 02:00:00', 350, 10), ('
19     '2023-05-01 03:00:00', 330, 11),
20     ('2023-05-01 06:00:00', 340, 11), ('
21     '2023-05-01 09:00:00', 350, 8),
22     ('2023-05-01 12:00:00', 70, 7), ('
23     '2023-05-01 16:00:00', 310, 12),
24     ('2023-05-01 19:00:00', 330, 14), ('
25     '2023-05-01 22:00:00', 340, 6),
26     ('2024-06-17 00:00:00', 290, 6), ('
27     '2024-06-17 03:00:00', 280, 4),
28     ('2024-06-17 06:00:00', 240, 3), ('
29     '2024-06-17 09:00:00', 210, 7),
30     ('2024-06-17 12:00:00', 220, 12), ('
31     '2024-06-17 14:00:00', 230, 14),
32     ('2024-06-17 17:00:00', 220, 14), ('
33     '2024-06-17 20:00:00', 210, 10),
34 ]
35 df_metar = pd.DataFrame(weather_data,
36     columns=['time', 'wind_dir', '
37     wind_speed'])
38 df_metar['time'] = pd.to_datetime(
39     df_metar['time'])
40 df_metar = df_metar.sort_values('time')
41 df_llegadas = df_llegadas.dropna(subset
42     =['time']).sort_values('time')
43 df_llegadas = pd.merge_asof(df_llegadas,
44     df_metar, on='time', direction='
45     backward')
46
47 # 2. SUAVIZADO DE GROUND SPEED
48 gs_mps_raw = np.sqrt(df_llegadas['vx'
49     ]**2 + df_llegadas['vy']**2)
50 df_llegadas['ground_speed_kts'] =
51     gs_mps_raw * 1.94384
52
53 df_llegadas['gs_smooth_kts'] =
54     df_llegadas.groupby('flight_id')['
55     ground_speed_kts'].transform(
56     lambda x: savgol_filter(x,
57     window_length=15, polyorder=2) if
58     len(x) > 15 else x
59 )
60
61 print("Step 6.2: Cálculo aerodinámico
62     (Gradiente de viento y TAS/IAS)...")
63
64 # 3. CÁLCULO DE VIENTO Y IAS
65 df_llegadas['aircraft_heading'] = (np.
66     degrees(np.arctan2(df_llegadas['vx'
67     ], df_llegadas['vy']))) % 360
68
69 angulo_relativo = np.radians(df_llegadas
70     ['wind_dir'] - df_llegadas['
71     aircraft_heading'])
72 df_llegadas['headwind_kts'] =
73     df_llegadas['wind_speed'] * 1.5 * np
74     .cos(angulo_relativo)
75 df_llegadas['tas_est_kts'] = df_llegadas
76     ['gs_smooth_kts'] + df_llegadas['
77     headwind_kts']
78 df_llegadas['ias_est_kts'] = df_llegadas
79     ['tas_est_kts'] / (1 + (0.02 * (
80     df_llegadas['altitud_ft_corr'] /
81     1000.0)))
82
83 print("Step 6.3: Extracción de VREF
84     Empírica (Data-Driven: 1000ft - 300
85     ft)...")
86
87 # 4. EXTRACCIÓN DE VREF DINÁMICA (Data
88     -Driven Estimation)
89 # Calculamos la mediana en el tramo de
90     aproximación final en lugar del
91     umbral de pista
92 tramo_estable = df_llegadas[(df_llegadas
93     ['altitud_ft_corr'] <= 1000) & (
94     df_llegadas['altitud_ft_corr'] >=
95     300)]
96 df_vref = tramo_estable.groupby('
97     flight_id')['ias_est_kts'].median().
98     reset_index()
99 df_vref.rename(columns={'ias_est_kts': '
100     dynamic_vref'}, inplace=True)
101
102 df_llegadas = pd.merge(df_llegadas,
103     df_vref, on='flight_id', how='left')
104
105 # 5. CRITERIOS OPERATIVOS (Adaptados a
106     los Tips del Documento)
107 # Tolerancia: +20 kts / -10 kts respecto
108     a la VREF extraída
109 limite_superior = df_llegadas['
110     ias_est_kts'] > (df_llegadas['
111     dynamic_vref'] + 20)
112 limite_inferior = df_llegadas['
113     ias_est_kts'] < (df_llegadas['
114     dynamic_vref'] - 10)
115
116 df_llegadas['speed_unstable'] =
117     limite_superior | limite_inferior
118
119 print("\nSuccess! VREF basada en mediana
120     y nuevos márgenes de tolerancia
121     aplicados.")

```


F Classification by aircraft family

```
1 diccionario_familias = {
2     'A320': 'Narrowbody', 'A319': 'Narrowbody', 'A321': 'Narrowbody', 'B738': '
    Narrowbody', 'B737': 'Narrowbody',
3     'B77W': 'Widebody', 'A332': 'Widebody', 'A339': 'Widebody', 'B789': 'Widebody',
    'B772': 'Widebody',
4     'E190': 'Regional', 'E195': 'Regional', 'AT76': 'Turboprop', 'AT75': 'Turboprop
    ',
5 }
6
7 if 'aircraft_type' in gate_snapshot.columns:
8     gate_snapshot['aircraft_family'] = gate_snapshot['aircraft_type'].map(
        diccionario_familias).fillna('Other')
9     print("--- INSTABILITY % PER AIRCRAFT FAMILY ---")
10    resumen_familias = gate_snapshot.groupby('aircraft_family')['global_unstable'].
        mean() * 100
11    print(resumen_familias.round(2).astype(str) + " %")
```

G Scoring function

```
1 # --- BLOCK 12: CONTINUOUS STABILITY
    SCORE (GAUSSIAN PENALTY) ---
2 import numpy as np
3
4 print("Step 12: Calculando el Stability
    Score (SS) con funci n Gaussiana
    ...\n")
5
6 # Aplicamos la f rmula:  $S = \exp(-x^2 / 2 \cdot \sigma^2)$ 
7 # x = desv o respecto al ideal
8 # sigma = tolerancia (cuanto mayor, m s
    permisivo)
9
10 # 1. Score de Velocidad (Target = VREF.
    Sigma = 10 kts)
11 desvio_vel = gate_snapshot['ias_est_kts']
    - gate_snapshot['dynamic_vref']
12 gate_snapshot['score_speed'] = np.exp(-(
    desvio_vel**2) / (2 * 10**2))
13
14 # 2. Score Lateral (Target = 0 . Sigma
    = 2.5 grados)
15 desvio_lat = gate_snapshot['
    track_deviation_deg']
16 gate_snapshot['score_lat'] = np.exp(-(
    desvio_lat**2) / (2 * 2.5**2))
17
18 # 3. Score de Descenso / Vertical Rate (
    Target = -750 fpm. Sigma = 250 fpm)
19 desvio_vr = gate_snapshot['savgol_vr_fpm
    '] - (-750)
20 gate_snapshot['score_vr'] = np.exp(-(
    desvio_vr**2) / (2 * 250**2))
21
22 # 4. GLOBAL STABILITY SCORE (Media de
    los 3, sobre 100 puntos)
23 gate_snapshot['stability_score_total'] =
    ((gate_snapshot['score_speed'] +
24
    gate_snapshot['score_lat'] +
25
    gate_snapshot['score_vr']) / 3) *
    100
26
27 print("    Scoring calculado para toda
    la flota en la puerta de 1000 ft.\n"
    )
28
29 # Mostramos la nota media por Familia de
    Aeronave (Esto es oro para el
    reporte)
30 if 'aircraft_family' in gate_snapshot.
    columns:
31     print("--- NOTA MEDIA DE ESTABILIDAD
        (SS) POR FAMILIA ---")
32     tabla_scores = gate_snapshot.groupby
        ('aircraft_family')['
        stability_score_total'].mean().round
        (1)
```

```

33     # Ordenamos de mejor nota a peor
34     nota
35     print(tabla_scores.sort_values(
36         ascending=False).astype(str) + " /
100")
35 else:
36     print("Nota Media Global de la Flota
:", round(gate_snapshot['
stability_score_total'].mean(), 1),
"/ 100")

```

Note: Part of the code has been omitted to comply with the length constraints of this report. However, the most important functions for the stability analysis can be found in this appendix.

References

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- [5] Zhenming Wang, Lance Sherry, and John Shortle. Feasibility of using historical flight track data to nowcast unstable approaches. In *2016 Integrated Communications Navigation and Surveillance (ICNS) Conference*. IEEE, 2016.
- [6] WeatherSpark. Weatherspark: Historical weather data and climatology. <https://weatherspark.com/>.